

Theorem (Markov's inequality). Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $X : \Omega \rightarrow [0, \infty]$ be a non-negative random variable. For any real number $a > 0$,

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a}.$$

1. Trivial and degenerate cases. If $X \equiv 0$ a.s., then $\mathbb{E}[X] = 0$ and $\mathbb{P}(X \geq a) = 0$ for every $a > 0$; the inequality holds with equality. Henceforth assume $a > 0$ and that X is an arbitrary non-negative random variable (not identically zero).

2. Indicator of the event $\{X \geq a\}$. Define the indicator function

$$\mathbf{1}_{\{X \geq a\}}(\omega) = \begin{cases} 1, & X(\omega) \geq a, \\ 0, & X(\omega) < a, \end{cases} \quad (\omega \in \Omega).$$

Since $\{X \geq a\} \in \mathcal{F}$, this function is measurable.

3. Pointwise bound. For every $\omega \in \Omega$ we have

$$\mathbf{1}_{\{X \geq a\}}(\omega) \leq \frac{X(\omega)}{a}. \quad (1)$$

Indeed, if $X(\omega) < a$ then the left-hand side is 0 $\leq X(\omega)/a$; if $X(\omega) \geq a$ then $X(\omega)/a \geq 1$, whence $1 \leq X(\omega)/a$.

4. Taking expectations. Because (1) holds pointwise, the monotonicity of the Lebesgue integral gives

$$\mathbb{E}[\mathbf{1}_{\{X \geq a\}}] \leq \mathbb{E}\left[\frac{X}{a}\right].$$

Linearity of the expectation allows us to pull out the constant $1/a$:

$$\mathbb{E}[\mathbf{1}_{\{X \geq a\}}] \leq \frac{1}{a} \mathbb{E}[X]. \quad (2)$$

5. Identification of the left-hand side. For any event $A \in \mathcal{F}$,

$$\mathbb{E}[\mathbf{1}_A] = \mathbb{P}(A).$$

With $A = \{X \geq a\}$ we obtain

$$\mathbb{E}[\mathbf{1}_{\{X \geq a\}}] = \mathbb{P}(X \geq a).$$

Substituting this into (2) yields the desired inequality

$$\boxed{\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a}}.$$

6. Equality conditions. Equality in (2) can occur only if every inequality used in the proof is an equality.

Proof. 1. **Equality in the pointwise bound (??).** This requires

$$\mathbf{1}_{\{X \geq a\}}(\omega) = \frac{X(\omega)}{a} \quad \text{for } \mathbb{P}\text{-a.e. } \omega. \quad (3)$$

2. **Solving (??).** If $X(\omega) < a$, then (??) forces $X(\omega) = 0$. If $X(\omega) \geq a$, then (??) forces $X(\omega) = a$. Hence (??) holds a.s. iff X takes only the two values 0 and a (with arbitrary probabilities). Formally,

$$\mathbb{P}(X \in \{0, a\}) = 1. \quad (4)$$

The degenerate case $X \equiv 0$ is included in (4) by allowing the probability of the value a to be zero.

Thus equality in Markov's inequality holds exactly when X is almost surely supported on $\{0, a\}$.

□